

BenGER Transformation: Exploring Automated Rubric Generation to Reduce LLM Grading Variance in German Legal Exams

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Large language models (LLMs) can grade answers to German law exams at scale. But the same answer gets different scores from different models, and even from one model on a second run. Some examiners grade with a grading sheet (*Bewertungsbogen*) that splits the expected solution into small steps worth fixed points. We ask whether a sheet that a model writes from the case and its reference solution makes LLM grades steadier, and at what cost to agreement with human graders. A first iteration on 15 exams and 45 human-graded answers used sheets from 12 models. On average the sheets did not make the stronger grading models (judges) steadier and tended to lower agreement with the human graders. With a sheet, several judges gave high scores to empty answers or to answers that only copy the case. In the rebuilt pipeline, one model writes small steps with explicit requirements, including paths for defensible alternative solutions and auxiliary opinions (*Hilfsgutachten*). The judge must back every point with a quote, and a fixed program checks that each quote stands in the answer. On the scripts of one police-law exam, graded on its author's own sheet, we test whether sheets make grades steadier and closer to the examiner's, which models write and use sheets well, and whether the pipeline meets German examination law. First runs on 15 scripts suggest that a sheet does not make repeated runs steadier, but brings the grades much closer to the examiner's. [PENDING: results of the full benchmark and the recommended pipeline]

CCS Concepts: • **Applied computing** → Law; Education; • **Computing methodologies** → Natural language processing.

Additional Key Words and Phrases: LLM-as-a-judge, grading rubrics, legal education, German law examinations, automated assessment, reliability and validity

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1 Introduction

German law students learn and are examined through case solutions (*Falllösungen*). A candidate receives a fact pattern (*Sachverhalt*) and writes a legal opinion in a prescribed appraisal style (*Gutachtenstil*). The state examinations grade such opinions on a statutory scale from 0 to 18 grade points [6]. Different examiners often give the same answer different grades [31]. Some correctors therefore first note what each part of the expected solution is worth [23, p. 330]. A few examiners make this binding in a grading sheet (*Bewertungsbogen*) [29, 61]. It splits the expected solution into small steps, each worth some assessment units (*Bewertungseinheiten*, BE), and a key maps the BE total to a grade. Appendix A lists the German terms and the technical terms we use.

Large language models (LLMs) can grade such texts at scale. An LLM used as a grader is called a *judge*. On German case solutions, a judge that applies the same 10 general criteria to every exam stays within the range of human disagreement, but its scores vary between judges and between

repeated runs [50]. For candidates this is a matter of equal opportunity (*Chancengleichheit*): comparable candidates must be graded by comparable standards [7, p. 52]. We test the examiners' remedy. One model, the *generator*, writes an exam-specific grading sheet from the case and its reference solution (*Musterlösung*), and the judge grades against it. The sheet also gives the examiner a document to read and correct.

A first iteration tested this remedy on 15 exams with 45 answers and blind human grades. On average the sheets did not make the stronger judges steadier, and they tended to lower agreement with the human graders. With a sheet, several judges gave high scores to empty answers or to answers that only copy the case (Section 6.1). We then rebuilt the pipeline from case to grade. We test it on the written answers (*scripts*) to a police-law exam set by one of us and graded on the author's own sheet and key, and we ask what German examination law requires of such grading.

- **RQ1 Variance.** Does a generated sheet make LLM grades steadier than grading without one, counting the scatter between runs, judges and the sheets one generator may write? Does it beat averaging several runs at equal cost?
- **RQ2 Validity.** How close are sheet-based grades to the examiner's and to a second human grader, and how many students would they wrongly pass or fail?
- **RQ3 Generators and judges.** Which models write sheets that yield grades close to the examiner's at low cost, and which judges use them well?
- **RQ4 Law.** Does sheet-based grading meet what German examination law asks of grading, on defensible (*vertretbar*) alternative solutions, follow-on errors (*Folgefehler*), reasons for each point and equal opportunity, and which uses does the evidence support?

Our contributions are:

1. A pipeline that turns doctrine into checkable structure (Section 4): steps with explicit requirements, separate paths for defensible alternative solutions, owed auxiliary opinions (*Hilfsgutachten*), and points only for quotes that a fixed program finds in the answer.
2. A study of all scripts of one exam that splits the scatter of grades into its sources (runs, judges, sheets) and compares sheets with averaging runs at equal cost (RQ1).
3. Agreement with the examiner, a second human grader and the examiner's own second grading, counting the students an LLM would wrongly pass or fail (RQ2), and a small benchmark of generators and judges (RQ3).
4. A legal analysis, co-written with the exam's author, that maps each requirement of examination law to a design feature and a test (Section 2, RQ4).

[DRAFT: redesigned 2026-09-27 after a simulated peer review; experiments E1 to E5 (Section 5.3) have not run in full]

[PENDING: one plain sentence per research question once E1 to E5 have run]

Most analyses are exploratory. One test is fixed in advance and will be published before the examiner grades the 23 held-back scripts. [PENDING: OSF registration link]

2 Grading practice in German law exams

[DRAFT: to be co-written with the exam author; every pending item below needs the author's input or experiments E2, E4 and E5]

The setting. The statutory scale has 7 grades (*Notenstufen*) from 0 to 18 grade points. 4 points (*ausreichend*, “sufficient”) is the lowest grade that still meets average requirements despite its flaws [6, § 1]. In practice a single exam is passed from this grade; the rules on passing are state (*Land*) law. Examiners set their own standards for an exam and must apply them to every script alike [9]. University exams usually have one grader [61, p. 29]. Some correctors first build a private point scheme [23, p. 330], and a few examiners make it binding in a sheet [29]. Such sheets are rare [61, p. 3]. Two principles bind any sheet. A defensible solution (*vertretbare Lösung*), consistently

Table 1. Three grading paradigms and our sheet. D2: the exam author’s sheet; Z2: Gregor Roth’s sheet and workbook (so far without case or scripts); partly fixed: a common practice that leaves a share of the BE to the grader (typical values; – = not set). ? = to be completed with the examiners. [PENDING: rows on alternatives and result items, confirmed with both examiners]

	Exam author (D2)	Roth (Z2)	Partly fixed	Our sheet
What is counted	absolute BE	% of the item above	BE per step, plus a free share	BE, allocated by code
Total	100 BE	100%	100 BE, 80% fixed	100 BE
Smallest items	46 steps	40 items, up to 4 levels deep	steps for the fixed share	steps with 1–6 requirements
How an item is scored	BE in half points	grade 0–18, whole points	BE per step; the free share at discretion	requirement met, partly or not; or step BE
Largest step	10 BE	9% of the grade	–	cap 10 BE
From items to total	sum of BE	weighted average of item grades	sum of BE	sum of BE
Pass mark	40% of BE	grade 4	the exam’s key	the exam’s key
Adjustment	?	manual bonus or malus	the free 20%	none
Credit for bare results	none; some titles	1 of 4 weighted	–	none; with its reasoning
Graders per script	one (see Section 5)	? (2 final-grade fields)	–	one LLM judge, run k times
Alternatives	?	?	–	requirements or forks

argued with weighty reasons, must not be marked as wrong. For exams that control access to a profession this follows from Art. 12(1) GG [7, p. 55]. And a follow-on error (*Folgefehler*) should cost points only once. A later step that correctly builds on the candidate’s own earlier decision still earns credit. [PENDING: exam author: source for the follow-on-error principle]

Three paradigms. We compare three ways of grading with our sheet (Table 1). The exam author’s sheet (D2) is a list of steps with absolute BE, grouped into sections. Gregor Roth’s sheet for a civil-law exam (Z2) is a tree. Each item carries a percentage of the item above it. Each of the 40 smallest items gets a grade from 0 to 18, and the final grade is their average, weighted by these percentages. This method needs no key [61, p. 12]. A third practice is common as well: about 80% of the BE go to steps, and the grader distributes the remaining 20% freely, for instance for structure or the overall impression. We do not adopt it: the free share is tied to nothing in the answer that a second grader or a judge could check, so it invites variance and cannot be reproduced. Our sheet follows the first paradigm. Every BE, in half points, belongs to a step. Each step lists the requirements that earn it, and code does all arithmetic (Section 4).

Q1: How should the BE be distributed? The author’s sheet spreads 100 BE over 46 steps with at most 10 BE per step. Some step weights changed between years (Section 5.2) [PENDING: exam author: why]. We took the step-size guidance of our generator (0.5 to 4 BE per step, at most 10 BE) from the two expert sheets, so step size is an input of our design, not a finding. E3 compares the BE distributions of generated sheets with the author’s sheet and E5 with Roth’s. [PENDING: BE distribution of the expert sheet and of generated sheets (E3)]

Q2: What earns credit, including method? In the author’s sheet every step credits an issue spotted or an argument made. No step credits a bare result; some step titles state a sub-result instead. Our generator follows the same rule. A result earns credit only with the reasoning that carries it. Where several results are defensible, the requirements credit the argument [7, p. 55]. The exam author credits method (structure, appraisal style, precise language) inside each section, not as a separate

block [61, p. 14]. [PENDING: exam author: confirm for D2] The first iteration kept style outside the points.

Q3: Where is the pass mark? The author’s key passes a script at 40% of the BE. The platform’s default key, which the first iteration used when grading without a sheet, passes at 50%. Where the pass mark lies is part of the examiner’s own evaluation. It cannot be fixed rigidly without regard to average results [7, p. 51], and once set it applies to every script alike [9]. The mark decides which students a judge passes or fails differently from the examiner. E2 sweeps it from 35% to 55% on stored outputs. [PENDING: E2]

Q4: Is the key linear or curved? The author’s key is piecewise linear with a kink at the pass mark: below it a grade point spans 10 BE, from the pass mark upwards 4 BE, up to grade 18 at 96 BE. The same shape, with passing costing more BE per grade than the top grades, is proposed in [61, p. 15]. [PENDING: exam author: confirm the reasons for D2] The platform’s default key has the same shape: below the pass mark a grade point spans 12.5 percentage points of the total on average, from the pass mark upwards 3.3. Roth’s paradigm needs no key: it averages statutory grades and passes from grade 4.

Q5: How are cardinal errors treated? A cardinal error (*Kardinalfehler*) is one that by itself can make a script fail. Examiners disagree on what counts, and the literature asks each sheet to define it in advance, with minus points or a grade cap [61, pp. 15 f.]. [PENDING: exam author: examples and expected loss for D2] Our sheet has no such cap yet. The follow-on-error rule and the standard of each requirement decide how far an early error carries (Section 4.3). E4 tests 5 fake answers with the losses the author expects.

3 Related work

Grading with LLMs. A judge reads an answer and returns a score. Such judges agree with human graders on average [43, 70] but are less dependable on a single answer. The same answer can get a different score when graded again [28, 57]. Scores change when answers are shown in another order [58, 64], and judges favour texts written by their own model [51]. A grader can also be steady and still wrong [32, 68]. Each use must therefore be checked against human graders [4, 10, 26]. Testing research splits score differences into parts due to the student, the grader and the occasion (generalizability theory) [5]. This has been applied to LLM graders [60, 65], and the wording of the grading prompt has been treated as one such part [49]. Most fixes change the grader, for example by pooling several models [63]. We change the sheet the grader uses, and we count the sheet a model happens to write as a source of differences of its own.

Grading sheets written by models. A rubric is a scoring guide that states what earns credit. A checklist breaks it into yes-or-no items. Such items can make LLM grading more consistent [11, 15, 36, 40, 69], and rubrics written for one question work better than general ones [33, 52, 62]. Human graders also agree more when they score an answer part by part than when they form one overall impression, at a cost in time [37]; for an LLM the time becomes computing cost. Models can write such rubrics. They derive weighted criteria from a reference answer [22, 27, 42] and point schemes for exam questions and proofs [25, 45, 46, 67]. Generated rubrics still fall short of expert ones [66, 71], and a regenerated marking scheme graded less accurately than the one the human graders had used [45]. Models that add up points themselves make mistakes [38, 54], so our generator states only relative weights and code allocates the points. Our sheet also encodes two rules of examination law: defensible alternative solutions, and follow-on errors that cost points only once.

Evidence and gaming. Attribution research asks whether a text is backed by the sources it cites [24, 56]. Some systems require exact quotes and check them in code [30, 35, 48]. Automated scoring has long been open to gaming by padding, nonsense or keyword lists [17, 55], and LLM graders

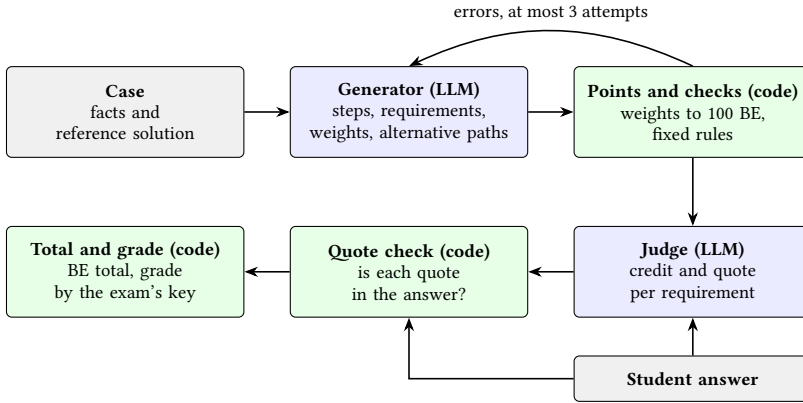


Fig. 1. The grading-sheet pipeline. Blue: model steps; green: code. The judge also receives the case facts and the reference solution.

can be steered by instructions hidden in an answer [12, 41]. Our judge must quote the answer for every credit. Code removes a credit whose quote is missing or turned into its opposite, and fake answers test every judge before use.

Grading law exams. Research first asked whether LLMs pass law exams [14, 34, 47] and now asks whether they can grade them. The graders receive model answers, marked passages, or rubrics written by experts or exam offices [1, 13, 16, 21, 53, 54, 59]. Others split answers into single legal points [18] or derive issue trees from court judgments [39]. German examiners disagree widely [31]. An examiner’s own point scheme cut the spread between human correctors by almost half [61], and LLM grades have been validated against the same examiner’s grades on the examiner’s own scheme [2]. To our knowledge, no study on law exams has a model write the point scheme from the reference solution, with scoreable paths for defensible alternatives and a follow-on-error standard per requirement, and tests the result against the examiner’s own sheet and grades.

4 The pipeline

[DRAFT: describes output specification checklist-4; the D2 runs of Section 6.2 used checklist-3 (fewer validator checks, a cap message without the advice to split); frozen after E0]

Figure 1 shows the pipeline. Models do two things only. The generator writes the sheet and says how much each part weighs. The judge grades an answer against the sheet and quotes the answer for every credit. Plain code, a fixed program and not a model, does everything that can be computed or checked. Models that add up points themselves make mistakes [38, 54], so no model ever states points.

4.1 Generation: the checklist sheet

The generator receives the case text, the reference solution and fixed instructions, the same for every model (Appendix O). It returns a *checklist sheet* with these parts:

- *Sections and steps.* Each step names one expected legal performance and carries a relative weight. The exam’s focal problems (*Klausurschwerpunkte*) are marked groups of steps. Each step belongs to a work result (*Arbeitsergebnis*), such as an opinion or a draft judgment.
- *Requirements.* Each step lists 1 to 6 requirements, each with a share of the step and a standard (*Maßstab*). A requirement is either correct (*richtig*), judged against the correct law, or consistent (*folgerichtig*), judged against the candidate’s own earlier decision. The latter carries

the follow-on-error rule. Each requirement names something the answer must write, which a quote can show. None may be met by leaving something out, none may be conditional, and none may ask only for a norm, a term or a result.

- *Forks and alternative paths.* A dispute that changes only a result or its reasons stays inside one step, and its requirement credits the argument, whatever the result. A structural fork (*Weichenstellung*) exists only where another defensible decision removes or adds steps. Each alternative path (*Lösungsweg*) has its own steps, its defensibility requirements (*Vertretbarkeit-sanforderungen*), and the same BE as the steps it replaces.
- *Auxiliary opinions (Hilfsgutachten).* A step can be marked as owed even when the candidate's own decision cuts it off, for instance when a contractual claim already fails at formation.

The model states weights, never points.

4.2 Points, checks and grades by code

Code turns the weights into 100 BE in half points. It uses the largest-remainder method known from seat allocation (Hamilton, in Germany Hare-Niemeyer) [3]. Each step first gets the whole half points its weight earns, and the leftover half points go to the steps with the largest remainders. Every step gets at least 0.5 BE, and a heavier step never gets fewer points than a lighter one. A requirement is worth its share of the step, and an alternative path is worth the steps it replaces. The first iteration's routine broke these rules in 11 of 170 sheets (Appendix D).

A program, the *validator*, then checks the sheet against fixed rules (Appendix J). Hard rules cover the structure, every cross-reference, the rules for requirements and forks, and the cap of 10 BE per step. If one fails, the generator gets its sheet back with the error messages and tries again, up to 3 times. Soft rules only log a warning. The rules compare words, so two steps that say the same thing in other words both pass. Every sheet records the hashes of the validator and of the instructions.

At the end, code adds up the credited points, rounds the total once to half points and maps it to a grade with the exam's key. Several runs can be averaged.

4.3 Judging

Code formats the sheet as text for the judge. The judge receives the case, the reference solution, the sheet and the answer, each clearly marked, and answers in a fixed format. We test two scoring units. In the *requirement* unit the judge marks each requirement as met, partly met or not met, which earns 1, 0.5 or 0 times its BE. In the *step* unit it awards each step up to its BE in half points, as on a classic sheet. Every credit needs a word-for-word quote from the answer. The judge follows fixed rules:

- *Placement.* Work counts for the step whose requirement it meets, even if it stands elsewhere in the same question and work result. It counts once. Work on another question counts only if the answer refers to it.
- *Follow-on errors.* Requirements with the consistent standard are judged on the answer's own earlier decisions. Other requirements are judged on the correct law, but an error that has already cost points is not charged again.
- *Auxiliary opinions.* Work on questions that the answer's own decision cut off earns credit like work in the main opinion. It never replaces the main result.
- *Alternatives.* The judge scores the steps of every path and names, with a quote, the path the answer follows. A path counts only if the answer gives the reasons its defensibility requirements demand. Code counts the named path (design A2); E4 compares two other designs (Appendix L).

Table 2. Data at a glance. A second grader grades 20 D2 scripts blind, 10 from each part, and the author grades 5 scripts a second time without seeing the first grade.

	What	Human grades	Role
D1	15 exams (civil, public and criminal law), each with an answer by a person, one by a person with AI help and one by a model	3 blind graders and the exam’s creator, out of 100, on the prior study’s 10 general criteria	development; first iteration (Section 6.1)
D2a	15 scripts of one police law exam (LMU), 2024–2025	one grader per script, step by step on the author’s sheet (grader of 9 to be confirmed)	development; first signals (Section 6.2)
D2b	23 scripts of the same exam, 2026	the author, blind [PENDING:]	held out: one registered test
Z2	Gregor Roth’s sheet and grading work-book for a civil-law exam	none	sheet comparison (E5)

- *Unforeseen paths.* If the answer takes a defensible path that the sheet does not foresee, the judge credits the corresponding step at equal depth of reasoning and flags the script for human review.
- *General rules.* Contradictory results without a decision fail the requirement on the result. Unnecessary alternative analyses earn nothing. Length, many citations and strings of keywords are not performance, and a short, correct application meets a requirement fully.

The judge is told to ignore instructions inside its inputs, and a probe tests this. It also writes short feedback for the candidate, which never changes points.

4.4 The quote check and its limits

Code checks that every quote really stands in the answer (Appendix I). It tolerates small differences in layout, inflection and typing. Words that change the meaning must match exactly: negations, result words such as *rechtmäßig* and *rechtswidrig* (lawful, unlawful), qualifiers and numbers. A quote also fails if a negation or a condition stands right next to it. A credit whose quote fails is set to zero.

The check has limits. It tests that a quote exists, not that it proves the requirement. A real sentence can be quoted for the wrong requirement. Correct work that the judge paraphrases earns nothing, and a sentence copied from the case text passes. The strict rules at the edges also reject some correct quotes. The first iteration had no quote check. [PENDING: regression replays at the freeze: quotes re-checked on D1, D2 and the stored D2b grades, and how many flip]

The experiments vary the generator, the judge, the scoring unit, the design for alternative paths and the number of averaged runs. The baselines are grading without a sheet, on the prior study’s 10 general criteria, and grading on the exam author’s own sheet. Options that the code supports but that we do not test here are listed in Appendix K.

5 Data and design

Table 2 gives the data at a glance.

5.1 D1: the first-iteration corpus

D1 holds the 15 exams of the BenGER corpus [50]. Three blind graders and the exam’s creator, who was not blind, graded every answer out of 100 points on the prior study’s 10 general criteria, the same criteria its LLM judge used. The blind graders often disagree: their highest and lowest grade for an answer lie 23.9 points apart on average. Section 6.1 compares with the mean of all 4

grades on the 30 answers written by people, as the prior study did. The new experiments use the blind grades as the main reference and the creator’s grade as a second one. The prior study’s judge was 10.8 points (2.7 grade points) from the mean human grade on average and ranked the answers much as the humans did ($r = .76$). D1 is our only data across many exams, but its reference is noisy.

5.2 D2: the expert exam

D2 is one police-law exam from LMU, set each year from 2024 to 2026. Its author is a co-author of this paper. The author’s sheet has 46 steps worth 100 BE in half points, and the author’s key passes a script at 40%. Some step weights changed between years, so the sheet exists in 3 versions with the same steps. The judge always sees the current one.

- D2a holds 15 typed scripts, 3 from 2024 and 12 from 2025, graded step by step on the sheet version then in use. We call these *reference grades*, because for 9 scripts the grader is still being confirmed.
- D2b holds 23 student scripts from 2026, written on our platform as a practice exam. We drop 2 uploads from the author’s own account (a test upload and a copy of a D2a script). The platform graded the scripts on submission with GPT-5.4 Mini, and the students saw these grades. The author now grades them blind, without seeing any model grade. [PENDING: blind expert grades for D2b]

D2 thus has 38 distinct scripts, or 37 if one pair that shares text counts once. We lack the 2024 case text, so the 3 scripts of 2024 are judged against a later one. [PENDING: the 2024 case text from the exam author]

All D2 texts were pseudonymized and checked for personal data before any research call to a model provider (Appendix N). No generator and no judge without a sheet sees the author’s sheet. A canary check confirms that the fixed instructions and examples in every prompt contain no terms from the author’s sheet or exam.

Z2 is Gregor Roth’s sheet and grading workbook for a civil-law exam, so far without case, reference solution or scripts. E5 generates sheets for this exam and compares them with Roth’s sheet. [PENDING: case and reference solution from Roth]

5.3 Experiments

We built the pipeline on D1 and D2a. D2b is held out. It holds new scripts of the same exam, not a new exam. The author grades them from exported scripts before seeing any generated sheet or model grade, and one registered test runs on them once. [PENDING: the registered hypothesis and its success criterion] Stability needs no human grade, so E1 uses all 38 D2 scripts. Table 3 lists the experiments. On the author’s sheet, which has no requirements, the judge scores per step; generated sheets are scored both ways, so that the sheet and the scoring unit can be told apart. The whole program costs USD 140 to 200 at list prices (Appendix M).

Measures. Accuracy is the average distance of a total from the human grade (mean absolute error, MAE), its sign (bias: too generous or too strict), the error in grade points on the 0 to 18 scale, and the students a model passes or fails differently from the human grade. For stability we split the spread of totals into real differences between scripts and noise from the judge, from the sheet a generator happens to write and from repeated runs [5]. Dependability, from 0 to 1, is the share of the spread that reflects real differences between scripts when a single run decides. We also count how often a second run lands on the other side of the pass mark [44] and compare sheets with averaging runs at equal cost. Each human reference was graded with one method, D1 on the 10 general criteria and D2 on the author’s sheet, so each favours the grading basis it shares.

Fake answers (probes). Before any benchmark run, every judge and sheet must clear a set of fake answers in each of 3 runs (Appendix H). Empty, copied, off-topic or manipulative answers

Table 3. Experiments of the extended study. [PENDING: actual costs per experiment]

Experiment	RQ	What we do
E0 Freeze and gate	RQ1, RQ2, RQ3, RQ4	Freeze one version of the pipeline. Test every judge and sheet with fake answers in 3 runs. Register the D2b test before any D2b grade exists.
E1 Steadier?	RQ1	Grade all 38 D2 scripts with no sheet, the author’s sheet and 10 generated sheets (2 models, 5 each, scored per requirement and per step): Luna 3 runs, DeepSeek-V4-Pro and GPT-5.4 Mini on part of them. Repeat on D1 (45 answers, 3 runs). Split the spread into its sources, count pass/fail flips between runs, compare with averaging runs at equal cost.
E2 Close to the examiner?	RQ2	Compare the E1 grades with the author (D2a; D2b as the registered test), a second grader (20 scripts, incl. the 9 with an unconfirmed grader) and the author’s second grading of 5 scripts. On D1, compare with the blind graders and test whether a model could replace one of them. Count wrong passes and fails at pass marks from 35 to 55%.
E3 Which models?	RQ3	5 generators with 3 sheets each on D2 and 5 D1 exams, one judge. 4 judges on fixed sheets, one of them an open model and one never used in development. Cost per valid sheet. The author reviews sheets without knowing which model wrote them.
E4 Doctrine and safety	RQ4	Fake answers the author writes, with the expected loss fixed in advance: a defensible other path, a consistent follow-on error, reversed decisive results, fluent but wrong application, padding, paraphrase, an answer written by a model, cardinal errors. Three designs for alternative paths. 200 quotes checked by hand.
E5 Second exam	RQ3, RQ4	Sheets generated for Roth’s exam, compared with Roth’s sheet (coverage, weights) and checked by Roth. Needs Roth’s case and reference solution.

must score about zero, and the reference solution must score high. Cutting one section of the reference solution must cost about its BE, and moving correct reasoning under another heading almost nothing.

Uncertainty. Apart from the registered test, we estimate and do not test. Every range is a 95% interval from redrawing exams (D1) or scripts (D2) many times (bootstrap). With 15 graded scripts only large differences are visible (Appendix G). Each judge runs at one fixed temperature in all comparisons. Scripts that the judge flags for review keep their totals in our analyses.

Development runs. Small runs on D1 and D2 and 4 automated review rounds (Section 11) shaped the pipeline. They used an earlier version. Section 6.2 reports the D2 runs as first signals; they do not enter the registered test. All runs of the extended study so far cost USD 34.17.

6 Results

Totals are points out of 100. A grade point is one step on the 0 to 18 scale. On the expert exam, 4 points make one grade point from the pass mark (40 points) up and 10 below it. r says whether two graders rank the scripts in the same order (1 = the same order); it ignores a constant shift. Brackets give 95% intervals. If the interval of a difference includes zero, we cannot tell whether the difference is real.

6.1 First iteration

The first iteration tested an earlier, richer sheet on D1. Each step carried a weight class, ordered levels of partial credit and follow-on-error rules. 12 generators wrote one sheet per exam, and a random draw picked one sheet per exam for all judges. 7 judges graded the 45 answers 3 times

Table 4. First iteration (D1): two judges with and without a generated sheet, in points out of 100 (about 3 points make one grade point above the pass mark). Run-to-run spread: standard deviation of a judge’s total over 3 runs on the same answer (45 answers). The other columns compare the first run with the mean of the four human grades (three blind graders and the exam’s creator) on the 30 answers written by people. Sheets written from an example: Appendix C.

Judge, grading basis	Run-to-run spread	Same order (r)	Average error	Too generous (+) / strict (–)
Luna, no sheet	2.3	.80	11.2	+4.3
Luna, generated sheet	2.0	.77	12.8	–6.9
GPT-5.4 Mini, no sheet	3.9	.75	11.2	–2.7
GPT-5.4 Mini, generated sheet	5.5	.41	19.4	+7.4

with that sheet (2 more once), and 6 of them also without a sheet, on the prior study’s 10 general criteria [50]. Totals are sums of the step scores (Appendix B).

Writing sheets worked. 170 of 180 generator–exam pairs yielded a valid sheet, and 10 of the 12 generators covered every exam. The sheets differed widely: the mean number of steps per generator ranged from 6 to 26 (Table 11).

The sheets did not make grading steadier (Table 4). Graded again, Luna’s total for the same answer moved by 2.3 points without a sheet and 2.0 with one (standard deviation over runs). Across the 5 judges other than GPT-5.4 Mini, the sheet changed this spread by +0.02 points on average [–0.63, +0.73], and Claude Sonnet 4.6 was steadier without a sheet (Table 13). With GPT-5.4 Mini included, the average would be +0.29. GPT-5.4 Mini’s own spread grew from 3.9 to 5.5 points, although its interval includes no change. The judges disagreed with one another far more than with themselves: the standard deviation of the 6 judges’ totals for the same answer was 7 to 10 points.

Nor did the sheets bring the grades closer to the humans. With the sheet, Luna ranked the answers about as before ($r = .77$ against .80) but turned from 4 points too generous to 7 points too strict. GPT-5.4 Mini’s average error rose from 11 to 19 points.

The sheets invited credit for nothing. With a sheet, Luna gave a near-empty answer about 96 of 100 points in 26 of 45 runs, and never did so without one. GPT-5.4 Mini gave empty answers a median of 95 points. Only Claude Opus 4.7 and DeepSeek-V4-Pro kept every fake answer near zero on every run (Table 15).

An automated review of the sheets by GPT-5.4 Mini did not predict which sheets graded well, and it favoured sheets that graded harshly (Table 14). The generated weights carried little signal, and big steps were not noisier per point than small ones (Appendices E and F). Appendix C gives the details.

We attribute these failures mainly to the first-iteration pipeline. It checked no quotes, listed the steps out of outline order, and gave the steps of alternative paths no fields of their own (Appendix B). The rebuilt pipeline (Section 4) addresses each defect, and E0 and E3 test again whether weaker judges can use a sheet safely.

6.2 First signals on the expert exam

These runs are a first look at the expert exam, not the benchmark. They used the pipeline as it stood (an earlier output specification), sheets from two generators and up to three judges. With 15 graded scripts, only differences in average error of about 2.5 to 8.4 points can be told apart from chance (80% power; Appendix G), so these are first signals, not a ranking. The yardstick is the reference grades of the D2a scripts. They were given on the examiner’s own sheet, which favours

Table 5. Expert exam: judge totals against the reference grades of the 15 D2a scripts, in points out of 100 (from the pass mark of 40 points up, 4 points make one grade point). Each value is computed per run and averaged over the runs. Grade points: the average error on the 0 to 18 scale under the exam’s key. Wrongly passed: students the judge passes whom the reference fails; wrongly failed: the reverse. Run to run: average difference between two runs on the same script. [†] not checked with the fake answers; [‡] failed that check (Table 7).

Grading basis	Judge	Runs	Average error [95%]	Grade points	Generous (+), strict (–)	<i>r</i>	Wrongly passed / failed	Run to run
No sheet	Luna [‡]	3	14.4 [10.2, 18.5]	3.3	+12.4	.84	5 / 0	1.8
No sheet	Mini [†]	1	13.1 [8.8, 17.8]	3.0	–2.8	.39	5 / 2	–
Examiner’s sheet, per step	Luna	3	5.1 [3.3, 7.0]	1.1	–3.4	.95	1 / 0.3	2.4
Examiner’s sheet, per step	DeepSeek	1	5.2 [3.2, 7.4]	0.9	–3.9	.95	0 / 1	–
Examiner’s sheet, per step	Mini [†]	1	9.4 [6.2, 13.0]	1.7	–1.6	.74	3 / 2	–
GPT-5.4 sheet, per requirement	Luna	3	6.1 [4.2, 8.2]	1.3	+2.2	.91	2.3 / 0	4.1
GPT-5.4 sheet, per requirement	DeepSeek	1	6.2 [4.1, 8.4]	1.4	+3.1	.91	3 / 0	–
GPT-5.4 sheet, per requirement	Mini [†]	1	10.3 [6.6, 14.4]	2.3	+5.7	.74	4 / 1	–
GPT-5.4 sheet, per step	Luna [†]	3	5.7 [3.8, 8.1]	1.3	–0.4	.89	2 / 0	4.6
GPT-5.4 Mini sheet 1, per requirement	Luna	3	7.3 [4.9, 10.1]	1.6	+6.1	.92	3 / 0	3.2
GPT-5.4 Mini sheet 1, per step	Luna [†]	3	8.3 [6.4, 10.2]	1.9	+5.6	.91	4.3 / 0	3.7
GPT-5.4 Mini sheet 2, per requirement	Luna [†]	3	7.3 [4.8, 10.0]	1.7	+7.2	.94	4 / 0	3.2
GPT-5.4 Mini sheet 2, per step	Luna [†]	3	6.9 [4.2, 9.7]	1.6	+5.9	.92	4 / 0	3.2

that sheet over generated ones, and for 9 scripts the grader is still being confirmed. The reference fails 5 of the 15 scripts.

Agreement with the reference grades (Table 5). Without a sheet, Luna was off by 14.4 points on average (3.3 grade points), almost always too generous (+12.4), and it passed 5 of the 5 students whom the reference fails. On the examiner’s sheet the same judge was off by 5.1 points (1.1 grade points), ranked the scripts almost as the reference did ($r = .95$); against no sheet, its error changed by $-9.3 [-14.5, -3.9]$ points. With generated sheets it was off by 5.7 to 8.3 points, mostly because they were more generous; every pass/fail disagreement was then a wrong pass. The GPT-5.4 sheet, scored per step like the examiner’s, came as close as the examiner’s own sheet (difference $+0.6 [-2.3, +3.7]$). Scoring per requirement instead of per step changed the error by at most 0.9 points. The platform’s default judge, GPT-5.4 Mini, was 9.4 points off even on the examiner’s sheet and passed or failed 5 of 15 students differently; without a sheet it ranked the scripts poorly ($r = .39$) and passed or failed 7 of them differently.

Stability. The sheets did not make repeated runs steadier (Table 6). Without a sheet, Luna’s total moved between runs by 2.9 points (standard deviation), on the examiner’s sheet by 3.4 and on generated sheets by 2.6 to 3.6. On D2a, the examiner’s sheet changed Luna’s run-to-run spread by $+0.6 [-0.5, +1.9]$ points against no sheet. The steadiness without a sheet came with a compressed scale, however: Luna spread the D2a scripts with a standard deviation of 8.0 points, the reference with 17.5, and it passed every D2a script, so a second run could not flip anyone. A grader can be steady and still wrong. A new sheet from the same generator moved a script’s total more than a new run did: one run with a fresh GPT-5.4 Mini sheet had a standard deviation of 4.0 to 4.1 points, and 6% to 16% of the scripts passed on one sheet and failed on the other. Averaging three runs without a sheet would lower the run-to-run spread to 1.7 points at about 1.9 US cents, but it cannot fix the generous level.

Fake answers. With every sheet, every judge we checked kept all six fake answers at zero (Table 7). Without a sheet, Luna gave them up to 14 points and failed the check; the judge without a sheet has no quote check. The rows of Table 5 that were not checked, or failed, are marked. Removing one part of the answer key cost about as many points as the sheet gives that part, and moving

Table 6. Expert exam, stability of GPT-5.6 Luna on all D2 scripts with at least two runs (no human grades needed), in points out of 100. Run to run: the standard deviation of a script’s total between runs. Between scripts: the spread of the scripts’ true totals. Dependability: the share of the spread that reflects the scripts when one run decides (0 to 1). Flips: how often a second run lands on the other side of the pass mark. A new sheet each time: one run with a freshly generated sheet, so the sheet adds its own noise; flips there compare two sheets.

Grading basis	Scripts	Run to run	Between scripts	Dependability	Flips	US cents per run
No sheet	38	2.9	9.6	.91	0%	0.6
Examiner’s sheet, per step	38	3.4	16.2	.96	9%	1.2
GPT-5.4 sheet, per requirement	38	2.9	15.2	.97	4%	1.6
GPT-5.4 sheet, per step	38	3.6	15.2	.95	5%	1.4
GPT-5.4 Mini sheet 1, per requirement	15	3.0	14.5	.96	9%	1.2
GPT-5.4 Mini sheet 1, per step	38	3.0	13.9	.96	2%	1.0
GPT-5.4 Mini sheet 2, per requirement	15	2.6	15.7	.97	0%	1.3
GPT-5.4 Mini sheet 2, per step	38	2.7	15.2	.97	4%	1.1
A new GPT-5.4 Mini sheet each time, per requirement	15	4.1	14.8	.93	16%	–
A new GPT-5.4 Mini sheet each time, per step	38	4.0	14.3	.93	6%	–

Table 7. Fake answers and edits of the answer key on the expert exam, in points out of 100. Fake answers: verdict on six fake answers (empty, the case text repeated, off-topic, off-topic from the same area, a keyword list, an instruction to award full marks); max: the highest score any of them received. The keyword list is built from a sheet’s terms, so it is not run without a sheet. Proposed: the most points the judge proposed for a fake answer before the quote check removed unsupported credit. Answer key: the reference solution graded as an answer. Part removed: points lost when one part of the answer key is deleted, against the points the sheet gives that part (matched by heading, a heuristic). Paragraph moved: points lost when a correct paragraph moves under another heading. Results reversed: points lost when the answer key’s results are negated.

Grading basis	Judge	Runs	Fake answers	Max	Proposed before quote check	Answer key	Part removed: lost / expected	Paragraph moved	Results reversed
No sheet	Luna	2	fail	14.0	13.0	100.0	21.0 / 0.0	2.5	37.5
Examiner’s sheet, per step	Luna	2	pass	0.0	23.0	99.5	19.5 / 21.0	1.3	18.3
Examiner’s sheet, per step	DeepSeek	1	pass	0.0	0.0	99.5	20.5 / 21.0	0.0	18.0
GPT-5.4 sheet, per requirement	Luna	2	pass	0.0	0.0	96.0	23.5 / 25.0	0.5	10.0
GPT-5.4 sheet, per requirement	DeepSeek	1	pass	0.0	0.0	94.5	21.0 / 25.0	0.0	5.0
GPT-5.4 Mini sheet 1, per requirement	Luna	2	pass	0.0	0.0	98.5	23.3 / 26.5	1.8	11.3
GPT-5.4 Mini sheet 1, per requirement	DeepSeek	1	pass	0.0	0.0	100.0	26.0 / 26.5	0.0	15.0

a correct paragraph under another heading cost almost nothing. On the examiner’s sheet, scored per step, Luna still proposed half a point for every step of an empty or off-topic answer, and only the rule that a credit needs a quote removed these points. Reversing the results of the answer key cost only 5 to 18 points with a sheet, but 38 without one: the requirements credit the reasoning whatever the result. What it should cost depends on how an examiner grades a conclusion that contradicts its own reasoning; E4 fixes this with the examiner in advance.

Generation. GPT-5.4 produced a valid sheet in 1 of 4 series of up to 3 attempts, GPT-5.4 Mini in 2 of 4. Of the 5 failed series, 3 ended on the cap of 10 BE per step, 1 on the rules for forks and 1 on an unreadable response. A generator states weights, not BE, and on an alternative path only splitting a step helps, which the error message did not yet say. A valid sheet cost USD 4.15 with GPT-5.4 and USD 0.23 with GPT-5.4 Mini, failed series included.

6.3 The redesigned experiments (E1 to E5)

[PENDING: E1 to E5]. E1 splits the spread of totals on all D2 scripts into differences between scripts and noise from runs, judges and the sheet a generator happens to write, gives the dependability of each grading basis and the share of pass/fail flips, and compares sheets with averaging runs at equal cost. E2 compares with the examiner, the second grader and the examiner's second grading and runs the registered D2b test. Every error is given in grade points and as students wrongly passed or failed. E3 compares 5 generators and 4 judges, E4 the doctrinal fake answers and the quote check, and E5 the generated sheets for a second exam against its examiner's sheet.

7 Discussion

[DRAFT: the answers per research question are written after E1 to E5]

What the first iteration showed. A generated grading sheet did not by itself make model grades steadier. Across the five stronger judges, the sheet changed the run-to-run spread by almost nothing on average, and one judge was steadier without it. The data cannot rule out a small effect either way. The sheets also exposed a worse failure. With a detailed sheet, several judges gave nearly full marks to answers that were empty or only repeated the facts of the case. GPT-5.6 Luna did so in 26 of 45 runs on a near-empty answer. An average agreement figure hides such failures. Every judge must therefore pass tests with fake answers before use, and code must check that each point rests on words that stand in the answer. Writing a sheet, grading with it and rating it are different skills. A model asked to rate sheets preferred those that graded harshly, so sheet writer and judge should be chosen by how close their grades come to human grades, not by how good a sheet looks.

What the expert exam suggests so far. On the examiner's own sheet, the best judge was about 5 of 100 points from the reference grades on average, or 1 grade point, and ranked the scripts almost as the reference did. Without a sheet the same judge was steady, but about 12 points too generous; it passed every student whom the reference fails and gave fake answers up to 14 points. Generated sheets came close to the examiner's sheet (6 to 8 points off), though mostly more generous. On this exam, then, a sheet did not make the judge steadier; it made its grades far more accurate. The platform's default judge was about 9 points off even on the examiner's sheet. A new sheet from the same generator moved a script's total more than a new run did, so a sheet should be fixed and approved once per exam. These are first readings on the scripts of one exam, and they favour the examiner's sheet (Section 8).

Answers to the research questions. [PENDING: one paragraph per research question after E1 to E5: first a plain answer (yes, no, partly, not yet), then what it means in grade points and in students wrongly passed or failed, then the number and its interval]

What this means for exam practice. Three points follow, all provisional. *Equal opportunity.* A newly generated sheet can move a script's total by several points. A sheet should therefore be generated once per exam, checked and approved by the examiner, and then used unchanged for every candidate. *Accuracy.* The generated sheets tended to be generous. In practice feedback a student may then believe they would pass when they would not. Such feedback should say so, or the sheet should first be calibrated on a few scripts the examiner has graded. *Reasons.* Each point comes with a quote that code has found in the answer, so an examiner or a student can see where it was earned. The code confirms only that the words are there, not that they meet the requirement. The quotes can support the examiner's reasons but cannot replace the examiner's own evaluation, which the examiner must make personally [8]. The evidence so far supports feedback on practice exams with these safeguards. A second reader that points the examiner to scripts where the totals differ is plausible but untested. It does not support grading exams that count towards a degree

without an examiner. EU law classes AI that evaluates learning outcomes as high-risk (AI Act, Annex III, point 3(b)), with the duties applying from December 2027 [19, 20]. Grading one script once costs about 1 to 17 US cents with the judges we priced (Appendix M). [PENDING: recommended pipeline and its cost per script]

8 Limitations

One expert exam. D2 is one case in one area of law, graded on its author’s sheet. D1 spans many exams, but its human reference is noisy. Results on D2 may not transfer to other exams, areas or examiners, and the second grader gives a human ceiling for one exam only.

Each yardstick favours one grading basis. The D2a reference grades were given on the examiner’s own sheet, and one setting grades with that same sheet. This favours it over generated sheets. On D1 all human graders used the 10 general criteria that the judge without a sheet also uses, which favours grading without a sheet. The second grader and the registered D2b test reduce this bias but do not remove it.

Open data questions. For 9 D2a scripts the grader is not yet confirmed, and the 3 scripts of 2024 are judged against a later year’s case text. The judge sees the current sheet also for scripts graded on an earlier version. The D2b students saw model grades, made with an earlier quote check, before the author grades their scripts; the author grades without seeing them.

The exam author is a co-author. The author wrote the exam, grades the scripts, reviews generated sheets and co-writes this paper, and so knows the study’s aims. Blind grading of D2b and the second grader will limit this. The author reviews generated sheets without knowing their source (E3), but this review is not independent.

Development on the test exam. During development, examples in the generator and judge prompts came from the D2 exam. We replaced them with neutral examples, and code now checks every prompt for words from the author’s sheet. It matches words only and cannot rule out that the prompts were shaped while the sheet was in hand. D2a is our development data, so results on it are optimistic.

The quote check. It confirms that a quote exists, not that it supports the credit (Section 4.4). It rejects some correct quotes and cannot catch a real sentence quoted for the wrong requirement.

Models and settings. GPT-5.6 Luna cannot be set to repeat its output. Even at its least random setting, DeepSeek-V4-Pro’s totals varied between runs (run-to-run spread 1.81 in the first iteration), so the spread mixes the model, its settings and the provider’s servers. Providers change and retire models; we record model ids and dates.

Exploratory design. Only the test on D2b is registered in advance, and it runs once. Everything else is exploratory. We report many comparisons without correcting for their number, and we added several first-iteration settings after seeing earlier results.

Scope. All D2 scripts were typed. Handwritten exams would need transcription, which we did not test. We did not test how examiners use a generated sheet or a model grade, for instance whether they correct it or simply accept it.

9 Ethics

Consent and personal data. The students whose D2 scripts we use consented to research use, for the 15 scripts of D2a and the 25 platform submissions of D2b. The consent does not extend to sending their personal data to model providers. All D2 texts are therefore pseudonymized and checked for personal data before any research call (Section 5.2). The platform grades that the D2b students saw were produced in the normal use of the platform, before this study. [PENDING: owner to confirm the wording on the platform-side grading and the providers’ data terms] D1 reuses the prior study’s corpus under the consent obtained there, with pseudonymized graders [50].

Roles. The exam author is a co-author, the source of the reference grades and the auditor of generated sheets. We report this as a limitation and mitigate it with blind grading and a second grader.

Students who saw LLM grades. The D2b students received LLM grades as feedback on a practice exam. [PENDING: owner: whether and how students learn the expert grades]

Use of the instrument. Generated sheets are studied as a research instrument for evaluating and supporting grading. They are not a replacement for human examiners in grade-bearing assessment, where legal and pedagogical safeguards apply.

Roth's sheet. Gregor Roth agreed to be named and to have his sheet and workbook used. We report only their structure, not their content.

10 Data and code availability

The platform is open source under the Apache 2.0 license at <https://github.com/SebastianNagl/benger-platform> [PENDING: tagged release or commit, archived with a DOI]. It contains the judge, the quote check and the aggregation. The generator's instructions and output specification, the validator, the allocation of points and the rendering of generated sheets belong to a proprietary extension of the platform. [PENDING: owner decision: release of the generator instructions, output specification and validator rules] The folder publications/Transformation holds the analysis scripts and every aggregate this paper reports. Every run records the hashes of its code, prompts and validator. Recomputing the aggregates needs data we do not release: the per-call records of D1, which contain third-party exam text, and the private D2 and Z2 material. D1's exam texts and the generated sheets, which restate the reference solutions, are not redistributed. For D2 and Z2 we publish only aggregates (Appendix N).

11 AI usage

This section covers AI tools used to do the research and write the paper, not the models under study. In the first iteration, the authors used OpenAI GPT-5.6 Sol to discuss approaches and Anthropic Claude Fable 5 to help write code and text. In the extended study, Anthropic Claude Opus 5.5 worked as a coding agent in the authors' repositories under the first author's direction. It wrote the rebuilt pipeline (output specification and validator, checklist judge, quote check), the experiment runner and all analysis scripts. It ran the experiments within budgets the first author approved one by one. It drafted most of the text, tables and appendices, including this section. It also searched the literature and checked the references it added against their sources. Further instances of the same model, given identical instructions, reviewed the code, the data handling, the statistics and the text in 4 rounds, simulated one peer review, and reviewed each section for readability and missing literature. The agent ran the scripts that extracted and pseudonymized the D2 texts on the authors' machine, so parts of the raw scripts may have reached Anthropic's service before pseudonymization. [PENDING: authors: confirm, and name the terms under which this happened] [PENDING: authors: who wrote Section 2 and the legal parts of the discussion] The authors checked [PENDING: every reference against its source; every number in the abstract, discussion and tables; the quote check and the allocation code]. The authors set the research questions and the design, supplied the exam, its grades and the sheets, decided every change of scope and spending, and take full responsibility for the paper.

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A Glossary

Table 8 glosses the German terms, Table 9 the technical terms.

Table 8. German terms used in this paper and their glosses.

Term	Gloss
<i>Falllösung</i>	Written case solution: the end-to-end work product that is graded.
<i>Sachverhalt</i>	Fact pattern: the facts of the case the candidate works on.
<i>Gutachtenstil</i>	Appraisal style: the conclusion follows the reasoning; the register graded here.
<i>Mustertlösung</i>	Expert reference solution against which submissions are graded.
<i>ausreichend</i>	“Sufficient”: lowest passing grade (4-6 of 18 points).
<i>Bewertungsbogen</i>	Grading sheet: the expected solution split into steps, each worth a number of BE.
<i>Bewertungseinheiten (BE)</i>	Assessment units: points on a grading sheet; 100 per exam here, in half points.
<i>Notenpunkte</i>	Grade points: the statutory scale from 0 to 18 of German law examinations.
<i>Notenstufen</i>	The seven named grades of that scale.
<i>Staatsexamen</i>	State examination: the examination that gives access to the legal professions.
<i>Chancengleichheit</i>	Equal opportunity: comparable candidates must be graded by comparable standards.
<i>Klausurschwerpunkte</i>	The exam’s focal problems.
<i>Maßstab</i>	The standard a requirement is judged by: <i>richtig</i> or <i>folgerichtig</i> .
<i>richtig</i>	Correct: judged against the correct law.
<i>folgerichtig</i>	Consistent: judged against the candidate’s own earlier decision.
<i>vertretbare Lösung</i>	Defensible solution: one argued with sound reasons, which must not be marked as wrong.
<i>Folgefehler</i>	Follow-on error: an error that later steps build on; it is not penalised twice.
<i>Weichenstellung</i>	Structural fork: a defensible decision that changes the further structure of the solution.
<i>Lösungsweg</i>	Alternative path after a structural fork, with its own steps.
<i>Vertretbarkeitsanforderungen</i>	The reasoning that makes an alternative path defensible.
<i>Hilfsgutachten</i>	Auxiliary opinion: the part a candidate still owes after a decision that ends the case.
<i>Arbeitsergebnis</i>	Work result, such as an opinion or a draft judgment.
<i>Kardinalfehler</i>	Cardinal error: an error grave enough to limit the grade as a whole.

Table 9. Technical terms used in this paper.

Term	Meaning
Judge	A language model that grades an answer.
Generator	A language model that writes a grading sheet from the case and its reference solution.
Code	A fixed program, not a model: it computes points, checks rules and quotes, and adds up totals.
Run	One grading of one answer. The same judge can grade the same answer several times.
Run-to-run spread	How far a judge’s total for the same answer moves between runs (standard deviation).
Average error (MAE)	The mean distance between a model’s total and the human grade, in points.
Bias	The mean signed distance: positive if the model is too generous, negative if too strict.
r	Whether two graders rank the answers in the same order (1: the same order); a constant shift does not count.
κ	Agreement on pass or fail, corrected for the agreement expected by chance.
95% interval	The range in which the true value lies with 95% confidence, from redrawing the exams or scripts many times.
Dependability	The share of the spread of totals that reflects real differences between scripts (0 to 1).
Fake answers (probes)	Made-up answers, such as an empty one or a copy of the case, that must score about zero.
Quote check	Code that tests whether each quote of the judge stands in the answer.
Validator	Code that checks a generated sheet against fixed rules and sends errors back to the generator.
Scoring unit	What the judge scores: each requirement (met, partly met, not met) or each step (points).

B Defects of the first-iteration pipeline

The first iteration’s pipeline had the defects below (Table 10). The rebuilt pipeline fixes the pipeline defects; the last two rows describe past runs, which stay as they were. Where a defect touches a first-iteration number, this paper reports the number without it, for instance totals as sums of the step scores.

Table 10. Defects of the first-iteration pipeline and how the rebuilt pipeline handles them.

Topic	Defect and handling
Apportionment	The routine broke the quota rule for 27 steps in 11 sheets, with 12 order inversions in 3. Of the 15 drawn sheets, 1 is affected (DeepSeek-V4-Pro; 8 of its 24 steps change by one point); all 11 affected sheets enter the generator ranking. Now floor-constrained Hamilton (Section 4.2).
Totals	The stored total was sometimes the model’s own, within half a point of the step sum. With step sums the zero-shot repeat SD moves for Luna from 1.97 to 1.96, for Mini from 5.52 to 5.51, and few-shot for Luna from 1.80 to 1.78. Now code computes every total.
Judge schema	Response keys followed storage order, not outline order. Now outline order (Section 4.3).
Alternatives	Steps of alternative paths had no response keys, and points above a step’s cap were dropped. Now every path has its own keys (Section 4.3). [PENDING: count of affected paths]
Verifier	The first iteration verified no quotes; the stored D2b grades used an earlier verifier.
Prompt wording	The generator prompt asked for exactly 100 points and also for relative weights; the weight class was rendered to the judge but unused. Now the model states weights only and code allocates.
Few-shot pool	The few-shot setting changed the generator pool along with the exemplar, so the two effects are confounded.

C First-iteration details

This appendix gives the details of the first iteration that Section 6.1 summarizes, and Tables 11 to 15 give its results per generator, judge and fake answer.

Generation (RQ3). 170 of 180 generator–exam cells hold a valid sheet. 10 generators covered every exam; Llama-4 Maverick covered 12 and GPT-5.4 Mini 8, mostly because feedback did not repair broken references of alternative paths. Of 220 dispatched series, 50 failed: 34 aborted at the provider (all GPT-5.4 Mini, credit errors), 14 failed validation and 2 failed parsing. First-attempt compliance over all series that reached the model spans 13% to 100%. Counted over valid sheets only, GPT-5.4 Mini looks better than it is: 38% there, but 17% over the series that reached the model (3 of 18) and 6% over all 52 of its series (Table 11). Generators did not converge: mean step counts range from 6.3 to 25.5, and stated weight sums from 34 to 179. A lexical screen found vague formulas that the prompt itself bans in 7 of 170 sheets (16 with near variants). The generation prompt asked for “exactly 100 raw points” in its first line, while its weighting rule said that no exact sum was needed because code normalizes; we disclose this as an ambiguity. The weight class was shown to the judge but played no part in the arithmetic.

Variance (RQ1). For Luna, the run-to-run SD falls from 2.25 (no sheet) to 1.96 (zero-shot sheet) and 1.78 (sheet written from an example, few-shot). The direction is consistent, but no single step is established: few-shot minus no sheet is -0.47 ($[-1.02, +0.07]$; sign test $p = .37$), zero-shot minus no sheet -0.29 ($[-0.77, +0.15]$; $p = .54$), and few-shot minus zero-shot -0.18 ($[-0.65, +0.24]$; $p = 1.00$). Across the 5 judges other than GPT-5.4 Mini that graded both ways, the zero-shot effect averages $+0.02$ ($[-0.63, +0.73]$; Table 13). This is the unweighted mean of the per-judge differences, without correction for multiple comparisons. Only Sonnet’s per-judge interval excludes zero, and it points the adverse way: Claude Sonnet 4.6 is steadier without a sheet ($+0.34, [+0.08, +0.64]$). For the weakest judge, GPT-5.4 Mini, the run-to-run SD rises from 3.87 to 5.51 ($+1.64, [-0.18, +3.45]$, clustered by exam; cell-level sign test $p = .02$, Wilcoxon $p = .007$). The steadiest few-shot sheets, a post-hoc subset by generator, reached 1.26 with Luna, and an in-sample floor (the best sheet per exam, chosen after the fact) 1.01; both are optimistic bounds without intervals. These contrasts use step sums; on the stored totals they differ by at most 0.02.

Agreement (RQ2). Against the mean human grade, Luna’s r moved from .80 to .77 and .70, and its bias flipped from $+4.3$ to -6.9 and -7.9 ; the few-shot exemplars had been chosen for stability alone. Its zero-shot MAE rose by $+1.6$ ($[-3.3, +7.0]$). GPT-5.4 Mini graded adequately without a sheet (r

= .75, MAE 11.2) but not with the sheet: $r = .41$, MAE 19.4, an increase of +8.2 ([+2.0, +15.8]). Few-shot sheets left it there ($r = .36$, MAE 21.3). Leave-one-exam-out bias correction shows that the error of the prior study's control judge was mostly a level shift (MAE 14.9 to 9.5), whereas Luna's zero-shot error was not (12.8 to 13.1).

Judges and probes. The 7 roster judges graded all answers against the drawn sheets (Table 12). Apart from Mini they reached $r .70$ to $.77$; Claude Opus 5 and GPT-5.6 Terra, one run each, $.74$ and $.75$. A set of 45 fake answers per type (the case text verbatim, a near-empty answer, another exam's reference solution) and each exam's own reference solution as a positive control tested execution (Table 15). Only Claude Opus 4.7 and DeepSeek-V4-Pro cleared every negative fake answer in every run. Luna credited the near-empty answer on 26 of 45 zero-shot runs and 24 of 45 few-shot runs, with 96.2 and 97.1 points on average when it did (median 100.0 and 99.5), but in 0 runs without a sheet. GPT-5.4 Mini failed 100 of 135 runs on negative fake answers with the sheet and 125 without a sheet, but its failures without a sheet were small (median 5 points on the empty answer) and its sheet failures near-total (median 95). Under the sheet, the off-topic answer scored at most 4.5 points with every judge except Mini (up to 94). All 7 judges passed the positive control under the sheet; Mini failed it without a sheet in 13 of 45 runs. On one answer that only restates the case, Mini under the sheet gave 99, 100, 100 points and Luna 0, 0, 0.

The first-iteration pipeline had no quote check and a generic system prompt. Its response schema listed the steps in storage order (by key length, then bytes), not in outline order, and the steps of alternative paths had no response keys, so a judge had to move credit for another path onto the replaced steps, where the per-step cap dropped any overflow [PENDING: number of alternative paths affected]. Budget neutrality thus held in the document, not in scoring. The new pipeline changes all of this at once, so E0 and E3 test the weaker judges again before we claim anything about them.

Generators and the audit (RQ3). A crossed sweep graded each exam's answers under every valid sheet of that exam: 170 sheets and 510 sheet-answer cells, Luna, 3 runs (Table 14). An automated audit by GPT-5.4 Mini rated each blinded sheet on 7 dimensions from 1 to 5. Within exams, the audit total shows no linear association with grading error ($r = .09$, $[-.08, .26]$; Spearman $.09$) and a weak one with stability ($r = -.13$, $[-.33, -.09]$; Spearman $-.22$). It correlates with the *absolute* bias ($r = .19$, $[.00, .35]$) and with the *signed* bias ($r = -.33$, $[-.46, -.18]$): higher-rated sheets grade harsher. At the generator level ($n = 12$), higher audit totals go with harsher grading (Spearman $-.70$ with Luna, $-.63$ with Sonnet): the Claude sheets top the audit (29.1 and 29.0 of 35) and grade among the harshest (bias -11.0 and -14.5). DeepSeek-V4-Flash sheets cut agreement ($r = .40$, the lowest) behind a mid-field audit (27.0). Llama-4 Maverick sheets made Luna least stable (repeat SD 4.07), but the audit ranked them lowest (22.3), so this failure was visible. GPT-5.4 Mini, the weakest judge and least compliant generator, ranked among the best sources on the exams where it produced a sheet (paired Δ MAE -3.2 on 8 exams), but its probability of ranking first is .47 and the estimate is conditional on survival. The ranking replicates moderately with Claude Sonnet 4.6 as judge (one run; Spearman $.50$; $.86$ among full-coverage generators). Few-shot generation raised Mini's valid sheets from 8 to 10 of 15; since the few-shot setting also used its own pool of 5 generators, its effects are confounded with that pool.

Weights and step size. The generated weights carry a small, judge-dependent signal (Appendix E). Re-aggregating the stored per-step scores, Luna keeps most of its agreement with class weights alone ($r .764$ vs $.769$) and loses some with uniform weights ($.724$). For DeepSeek-V4-Pro, class weights do worse than uniform ones ($.683$ vs $.691$). Uniform weights lowered the run-to-run SD for 6 of 7 judges, and the actual weights beat 75 to 99% of within-sheet permutations. The judges saw the actual step maxima, so these re-aggregations are conditional on them. Big steps are not noisier per point (Appendix F). Steps of 15+ points held 22% of the points and, for most judges, a larger

Table 11. First iteration, generation by generator: valid sheets, first-attempt compliance over successful sheets and over all series that reached the model, mean attempts per valid sheet, mean step count, and automated-audit total (of 35).

Generator	Valid	1st (valid)	1st (series)	Attempts	Steps	Audit
Claude Opus 4.7	15/15	73%	73%	1.27	17.1	29.1
Claude Sonnet 4.6	15/15	13%	13%	1.93	22.0	29.0
DeepSeek-V4-Flash	15/15	80%	71%	1.27	18.5	27.0
DeepSeek-V4-Pro	15/15	100%	100%	1.00	25.5	26.6
GPT-5.4	15/15	100%	100%	1.00	16.8	28.5
Gemini 3.1 Flash-Lite	15/15	80%	80%	1.20	6.3	24.4
Gemini 3.1 Pro	15/15	100%	100%	1.00	15.3	26.5
Qwen3-235B-Thinking	15/15	40%	40%	1.60	12.8	25.6
Qwen3.5-122B	15/15	13%	13%	1.93	9.3	26.0
Qwen3.6-35B	15/15	60%	56%	1.60	11.5	26.3
Llama-4 Maverick	12/15	75%	60%	1.25	6.5	22.3
GPT-5.4 Mini	8/15	38%	17%	1.88	17.1	27.4

Table 12. First iteration, judges on the drawn sheets (step sums), sorted by MAE. r and MAE: first run against the mean human grade (30 answers). SD_{rep} : run-to-run SD over 3 runs on the listed number of answers. Battery: fail(N) counts failing runs on negative fake answers. P01: the first run on an answer that only restates the case. Below the line: judges with one run, from the database extraction.

Judge	r	MAE	SD_{rep}	Answers	Battery	Pos.	P01
GPT-5.6 Luna	.77	12.8	1.96	45	fail(32)	pass	0
Qwen3.5-397B	.74	13.2	5.43	45	fail(19)	pass	0
Gemini 3.1 Pro	.73	13.6	2.12	44	fail(20)	pass	0
Claude Sonnet 4.6	.77	15.2	1.17	45	fail(6)	pass	0
Claude Opus 4.7	.74	15.5	1.25	36	pass	pass	0
GPT-5.4 Mini	.41	19.4	5.51	45	fail(100)	pass	99
DeepSeek-V4-Pro	.70	19.7	1.81	45	pass	pass	0
Claude Opus 5	.74	14.1	–	–	–	–	0
GPT-5.6 Terra	.75	14.4	–	–	–	–	0

share of the summed per-step variance (Luna 36%). But if every step had the same relative noise they would hold 52%, because variance grows with the square of a step's points; only DeepSeek exceeded that, and Luna's relative SD on big steps (0.037) was no higher than on steps of 3 to 5 points (0.042). The case for small steps is averaging: independent errors on many small steps partly cancel, while the error on one big step enters the total whole. Step count correlated only weakly with error ($r = .14$) and with repeat SD ($-.14$). The new generator writes small steps.

Table 13. First iteration, run-to-run SD of each judge with the zero-shot sheet and without a sheet on the same 45 answers, 3 runs each (step sums). 95% intervals from a bootstrap that redraws exams; sign test on answers. The pooled estimate averages the judges other than GPT-5.4 Mini without weights; no correction for multiple comparisons.

Judge	Sheet	No sheet	Δ	95% interval	Sign p	Pooled
GPT-5.6 Luna	1.96	2.25	-0.29	-0.77 to +0.15	.54	yes
Claude Sonnet 4.6	1.17	0.83	+0.34	+0.08 to +0.64	.64	yes
Gemini 3.1 Pro	2.14	2.42	-0.28	-0.83 to +0.28	.88	yes
DeepSeek-V4-Pro	1.81	1.67	+0.13	-0.48 to +0.65	1.00	yes
Qwen3.5-397B	5.43	5.22	+0.21	-3.09 to +3.80	.37	yes
GPT-5.4 Mini	5.51	3.87	+1.64	-0.18 to +3.45	.02	no
Pooled (unweighted)			+0.02	-0.63 to +0.73		

Table 14. First iteration, sheet sources ranked by outcome with Luna as judge: agreement, MAE, signed bias, run-to-run SD, and the within-exam paired Δ MAE (negative: better than the exam’s average source), with the bootstrap median rank and its 95% interval. Right: MAE and bias with Claude Sonnet 4.6 as judge, one run.

Source	Sheets	r	MAE	Bias	SD _{rep}	Δ MAE	Rank	MAE _s	Bias _s
Gemini 3.1 Flash-Lite	15	.82	11.4	+2.0	2.40	-2.4	2 [1–10]	13.7	-5.9
Qwen3-235B-Thinking	15	.81	12.0	-5.5	2.19	-1.9	3 [1–8]	13.2	-11.2
Qwen3.6-35B	15	.78	12.5	-4.9	2.21	-1.4	4 [1–8]	13.8	-10.2
GPT-5.4	15	.80	12.6	-6.2	1.80	-1.2	5 [2–8]	14.7	-10.8
Qwen3.5-122B	15	.76	12.8	+1.3	1.73	-1.1	5 [1–10]	14.1	-5.8
Gemini 3.1 Pro	15	.75	14.4	-9.0	1.97	+0.6	8 [2–12]	16.0	-13.1
Claude Opus 4.7	15	.79	14.7	-11.0	1.86	+0.9	8 [4–11]	17.7	-17.0
Claude Sonnet 4.6	15	.81	15.5	-14.5	1.70	+1.7	10 [5–12]	17.8	-17.4
DeepSeek-V4-Pro	15	.79	15.6	-11.6	1.93	+1.8	10 [7–12]	18.7	-16.4
DeepSeek-V4-Flash	15	.40	16.4	-3.5	3.18	+2.6	11 [4–12]	16.0	-12.2
<i>Incomplete coverage (conditional on a valid sheet):</i>									
GPT-5.4 Mini	8	.88	10.3	-0.5	1.66	-3.2	2 [1–8]	12.4	-5.6
Llama-4 Maverick	12	.78	15.9	+9.8	4.07	+2.7	11 [5–12]	11.8	+1.0

Table 15. First iteration, fake answers by setting and judge: verdict on the negative fake answers (fail(N) = failing runs), positive control, and the median/p95/max score per type. Criteria in every run: repetition < 5, empty = 0, off-topic $\leq$ 10, reference solution $\geq$ 70.

Setting, judge	Battery	Pos.	Repetition	Empty	Off-topic	Model sol.
Zero-shot, Claude Opus 4.7	pass	pass	0/0/0	0/0/0	0/1/4	99/100/100
Zero-shot, Claude Sonnet 4.6	fail(6)	pass	0/100/100	0/24/81	0/0/0	100/100/100
Zero-shot, DeepSeek-V4-Pro	pass	pass	0/0/0	0/0/0	0/0/0	100/100/100
Zero-shot, GPT-5.4 Mini	fail(100)	pass	100/100/100	95/100/100	8/83/94	100/100/100
Zero-shot, GPT-5.6 Luna	fail(32)	pass	0/100/100	89/100/100	0/1/5	100/100/100
Zero-shot, Gemini 3.1 Pro	fail(20)	pass	80/100/100	100/100/100	0/0/0	100/100/100
Zero-shot, Qwen3.5-397B	fail(19)	pass	100/100/100	79/100/100	0/0/0	100/100/100
No sheet, GPT-5.4 Mini	fail(125)	fail(13)	45/69/71	5/5/7	13/21/22	82/91/95
No sheet, GPT-5.6 Luna	fail(17)	pass	0/2/93	0/0/0	8/18/19	98/100/100
Few-shot, GPT-5.4 Mini	fail(96)	-	99/100/100	95/100/100	5/89/95	-
Few-shot, GPT-5.6 Luna	fail(31)	-	0/100/100	90/100/100	0/0/0	-

D Apportionment audit

The audit re-derives the points of all 170 first-iteration sheets (2542 steps) from their stated weights under each routine of Table 16. A step lies outside its fair range when its points differ from its quota

Table 16. Apportionment of the 170 first-iteration sheets under three routines: steps outside their fair range, sheets affected, order-inversion pairs, pairs of distinct weights that receive equal points, median relative rounding error, and steps whose points differ from the as-run result.

Routine	Steps	Sheets	Inversions	Collapsed	Rel. error	Changed
As run (first iteration)	27	11	12	458	0.033	0
Hamilton, floor first (integer)	0	0	0	286	0.033	54
Hamilton, floor first (half points)	0	0	0	0	0.020	1060

by a full unit or more; an order inversion is a pair in which the lighter step receives more points. The as-run routine is reproduced exactly by a copy of the original code.

E Weight sensitivity

Every first-iteration judgment stores a score and a maximum per step. Re-weighting the stored fractions shows how much the generated allocation contributes to agreement and stability without new judge calls (Table 17). *Actual* reproduces the step sums; *class* weights each step by its weight class (routine 1, load-bearing 2, focal 3); *uniform* counts every step the same; *permuted* shuffles a sheet’s own points across its steps. The judges saw the actual maxima, so every scheme is conditional on them.

Table 17. Weight sensitivity of the first-iteration sheets: agreement r with the human pool under five weight schemes, the share of within-sheet permutations whose r falls below the actual one, and repeat SD under the actual and uniform weights.

Judge	Actual	Model	Class	Uniform	$\sqrt{\text{pts}}$	Perm.	SD act.	SD unif.
Claude Opus 4.7	.741	.738	.730	.724	.738	88%	1.25	1.15
Claude Sonnet 4.6	.772	.771	.765	.758	.769	87%	1.17	1.32
DeepSeek-V4-Pro	.700	.697	.683	.691	.699	75%	1.81	1.61
GPT-5.4 Mini	.411	.410	.383	.378	.398	92%	5.51	4.85
GPT-5.6 Luna	.769	.767	.764	.724	.754	99%	1.96	1.77
Gemini 3.1 Pro	.732	.728	.724	.698	.722	96%	2.12	1.90
Qwen3.5-397B	.745	.742	.731	.716	.738	93%	5.43	5.05

F Step size and variance

For every judge with at least 3 runs per answer, we sum the per-step variance across runs by step size and compare it with the share of points in that size class (Table 18). The share ignores covariance between steps, so it is a share of *summed* step variance. The right comparison is the share of *squared* points: if every step had the same relative noise (SD proportional to its points), a size class would hold exactly that share of the variance. The relative SD (SD over points) of big steps and of steps of 3 to 5 points tests this directly.

Table 18. Steps of 15+ points in the first-iteration sheets: answers and step cells, share of points, share of squared points (the variance share under equal relative noise), share of summed per-step repeat variance, and the relative SD of these steps and of steps of 3 to 5 points.

Judge	Answers	Step cells	Points	Points ²	Variance	Rel. SD	Rel. SD 3–5
DeepSeek-V4-Pro	45	45	22.4%	52.1%	65.0%	0.028	0.018
Claude Opus 4.7	36	21	15.0%	44.6%	40.7%	0.025	0.024
Gemini 3.1 Pro	44	44	22.6%	52.7%	42.7%	0.036	0.047
GPT-5.6 Luna	45	45	22.4%	52.1%	36.0%	0.037	0.042
Claude Sonnet 4.6	45	45	22.4%	52.1%	32.5%	0.015	0.021
GPT-5.4 Mini	45	45	22.4%	52.1%	28.4%	0.061	0.088
Qwen3.5-397B	45	45	22.4%	52.1%	25.5%	0.041	0.082

G Precision

This appendix shows how large a difference must be before the data can reveal it. The paper uses $\alpha = .05$ throughout; the column for $\alpha = .025$ is kept for comparison. The minimum detectable effect of a paired contrast is $(t_{1-\alpha/2, n-1} + t_{\text{power}, n-1}) \sigma_d / \sqrt{n}$. We take σ_d from the closest existing contrast, Luna with and without a generated sheet on D1, so the values are planning proxies (Table 19). The D2 proxy uses the D1 human reference, which is noisier than step-by-step expert grades; the first D2a runs show smaller spreads, and Section 6.2 reports the detectable differences they imply.

Table 19. Minimum detectable effects at power .8 (t -based, paired). σ_d : SD of the paired differences in the proxy contrast.

Contrast	n	σ_d	$\alpha = .025$	$\alpha = .05$
Repeat SD, D1	45	1.87	0.88	0.80
Repeat SD, D1, 5 runs (optimistic)	45		0.63	0.57
Absolute error, D2a (15 scripts)		12.4	10.9	9.7
Absolute error, all distinct D2 scripts (38)		12.4	6.4	5.8
Absolute error, D2, dependent pair once (37)		12.4	6.5	5.9

H Probe battery

Before a model may grade, it must clear these fake answers with every sheet it will use, in each of 3 runs (Table 20). A judge that fails a criterion in any run is not used with that sheet. E4 adds fake answers that the exam author writes, among them the reference solution rewritten along one alternative path.

Table 20. Fake answers (probes) of the extended study. Totals are out of 100 BE.

Fake answer	Input	Criterion
empty	a near-empty answer	total = 0
repetition	the case text, verbatim	total < 5
off-topic	the reference solution of another exam	total $\leq$ 10
same-area off-topic	the reference solution of another exam in the same area	total $\leq$ 10
keyword list	norms and key terms of the sheet, no sentences	total $\leq$ 10
injection	an answer that instructs the judge to award full marks	total = 0
reference solution	the exam's own reference solution	total $\geq$ 70
section removed	the reference solution with one section removed	drop close to the removed section's BE
paragraph moved	correct reasoning moved under another heading	drop close to 0
results reversed	result sentences of the reference solution negated	reported, no threshold

I Verifier rules

The verifier (`services/workers/ml_evaluation/llm_judge_evaluator.py`) accepts a quote only if every fragment passes:

- *Normalization*. Unicode normalization; literal line-break escapes, markup escapes, footnote marks, emphasis markers and inline HTML removed; quotation marks and dashes unified; the result marks “(+)” and “(-)” turned into the words *positiv* and *negativ* (positive, negative); case folded; whitespace collapsed.
- *Fragments*. A quote is split at ellipses. Each fragment needs at least 3 tokens, or two tokens with 15 letters together, and more than a norm citation. The whole quote needs 4 word characters.
- *Matching*. Fragments with fewer than 4 tokens must match on word boundaries. Longer ones may match as a substring or token by token in order, with up to 2 skipped answer words between quoted words and one missed quoted word per 8. Unprotected words of at least 5 letters may differ by one edit or a small inflection. Glued passages without an ellipsis are checked sentence by sentence.
- *Protected tokens*. Negations, result words and their antonyms (including every *un-* form), qualifiers such as *nur* (only) and *teilweise* (partly), Roman numerals and every token with a digit must match exactly. They may not be dropped, added, swapped or stepped over. Numbers never match part of a longer number.
- *Edges*. A match fails if a negation or a condition (*ob*, *wenn*, *falls*, *sofern*, *soweit*: whether, if, in case, provided, insofar) stands right before it in the same clause, or a negation, a negative result mark or a condition right after it in the same sentence.

J Validator checks

The validator checks every generated sheet before it is used. *Hard* checks fail an attempt and return their messages to the model: the document structure and every cross-reference; steps above 10 BE; steps above 3 BE with a single requirement; requirements that can be met by an absence or that are conditional; steps named as a result (except the finding after a taking of evidence and the operative part of a judgment, *Tenor*); regular steps that depend on a step of an alternative path; forks whose steps depend on replaced steps; a fork that replaces a whole section; and a path whose requirements repeat those of the replaced steps (word overlap of at least 0.6). *Soft* checks only log hints: broader absence patterns; forks that replace exactly one step with one step; forks above 20

BE; near-duplicate requirements; requirements that only name something and are shorter than 8 words; a sheet on which most steps exceed 4 BE; and rescaled sections in top-down allocation.

K Pipeline options not tested here

The code supports the options below. The experiments in this paper do not vary them.

- *Weight elicitation.* The generator can state flat weights over all steps; shares per level, where each item states its percentage of the item above it (Roth's paradigm, Z2); or section totals first and then the steps inside (the D2 paradigm). Code normalizes all three. Every sheet in this paper uses flat weights.
- *Case preparation.* Code removes markup escapes and tables of contents from the case text and the reference solution and keeps footnotes.
- *Allocation detail.* Steps whose share falls below the minimum are set to it before the remainders are ranked, and the rest is apportioned again. Every other step then lands within one half point of its exact share, and weights that already sum to the total pass unchanged.
- *Merged view.* A second rendering merges consecutive steps of a section until a merged step reaches about 10 BE, without crossing a focal group, a fork or an auxiliary-opinion marker. Totals do not change.
- *Rating unit.* The judge grades each step from 0 to 18 against the statutory grade definitions. A step grade maps back to BE at the middle of its band under the exam's key, with 18 points worth the full step. The weighted average of the step grades, Roth's aggregation, is reported next to it.
- *Two-judge mean.* The totals of two judges can be averaged.
- *Shown step scores.* The step scores shown to a reader are distributed from the rounded total by largest remainder, so that they add up to it.

L Designs for alternative solution paths

A dispute that changes only a result or its reasons (a local alternative) stays inside a step as an outcome-neutral requirement, such as “discusses the dispute with the arguments of both sides; both results defensible”. A structural fork replaces named steps of the reference solution's path with the steps of each alternative path, within the same budget. We compare three designs (E4):

- *A0, replace.* Only the reference solution's path is scored. The rendered sheet tells the judge to credit work on another defensible path on the replaced steps. This is the first iteration's design; it needs the step or rating unit, because a requirement of a replaced step describes that step's own requirement.
- *A1, branch, best path.* The judge scores the steps of every path; code counts the path with the most points.
- *A2, branch, declared path.* As A1, but code counts the path the judge declares the answer follows, backed by a verified quote; without one it counts the reference solution's path.

A1 and A2 come from the same call. A path counts as followed only if the answer argues it as its defensibility requirements demand; a bare other result does not count.

M Costs

The program is planned at USD 140 to USD 200 (Table 3).

Table 21. Planned cost in USD per judge call (about 25k input and 6k output tokens) and per generated sheet, from pilot-measured tokens at catalog prices. Open-model sheets cost less than the value shown. [PENDING: measured costs]

Model	USD
GPT-5.6 Luna	0.013
GPT-5.4 Mini	0.046
DeepSeek-V4-Pro	0.070
Gemini 3.1 Pro	0.120
GPT-5.4	0.150
Claude Sonnet 4.6	0.170
Sheet: open models	0.05
Sheet: GPT-5.4 Mini	0.15
Sheet: GPT-5.4	0.60
Sheet: Claude Opus 4.7	1.00

N Data statement

D1. The first-iteration corpus of 15 German exams and 45 answers with blind human grades [50]. Exam texts are third-party material used under the permissions documented for that corpus and are not redistributed. Human grades are pseudonymized.

D2. One police-law exam with its author’s sheet and key, 15 graded scripts (D2a) and 23 student submissions to a practice exam (D2b), used with the students’ consent for research. All texts are pseudonymized before any research call to a model provider. The check found and removed 1 line with a student’s name and 1 examiner header; 9 grade spreadsheets carried a private e-mail address, which no derived file contains. The material stays private; only aggregate statistics are published. [PENDING: who wrote the scripts and when, exam status, consent process, storage location and deletion date]

Z2. Gregor Roth’s grading sheet and workbook for a civil-law exam, without case, solution or scripts. Private; described only by its structure.

Model calls. Every call records the model id, parameters, prompt hashes and the code version. For D1 we release aggregates; the per-call records are withheld for size. [PENDING: owner: release the per-call records with the dataset release?]

O Output specification of the checklist sheet

[PENDING: the verbatim output specification (German) and its English translation, after the E0 freeze]

P Example generated sheet

[PENDING: one generated checklist sheet for a D1 exam, as rendered for the judge, after the E0 freeze; D1 only, never D2]